The TPS collects data continuously, typically in *real time*—that is, as soon as the data are generated—and it provides the input data for the corporate databases. TPSs are considered critical to the success of any enterprise because they support core operations. Significantly, nearly all ERP systems are also TPSs, but not all TPSs are ERP systems. In fact, modern ERP systems incorporate many functions that previously were handled by the organization's functional area information systems. You study both TPSs and ERP systems in detail in Chapter 10.

ERP systems and TPSs function primarily within a single organization. Information systems that connect two or more organizations are referred to as **interorganizational information systems (IOSs)**. IOSs support many interorganizational operations, of which *supply chain management* is the best known. An organization's **supply chain** is the flow of materials, information, money, and services from suppliers of raw materials through factories and warehouses to the end customers.

Note that the supply chain in Figure 1.5 shows physical flows, information flows, and financial flows. Digitizable products are those that can be represented in electronic form, such as music and software. Information flows, financial flows, and digitizable products go through the Internet, whereas physical products are shipped. For example, when you order a computer from **www.dell.com**, your information goes to Dell via the Internet. When your transaction is completed (that is, your credit card is approved and your order is processed), Dell ships your computer to you. (We discuss supply chains in more detail in Chapter 11.)

Electronic commerce (e-commerce) systems are another type of interorganizational information system. These systems enable organizations to conduct transactions, called business-to-business (B2B) electronic commerce, and customers to conduct transactions with businesses, called business-to-consumer (B2C) electronic commerce. Figure 1.5 illustrates B2B and B2C electronic commerce. Electronic commerce systems are so important that we discuss them in detail in Chapter 7, with additional examples interspersed throughout the text. IT's About Business 1.2 shows how various technologies have enabled Lemonade Insurance to grow rapidly via e-commerce.

IT's About Business 1.2

MIS POM Digital Transformation at Hero MotoCorp Ltd.

Digital transformation has brought a paradigm shift in the way organizations create and deliver value. Organizations in every sector are now exploring ways to use new technologies like cloud, AI, IoT to profoundly transform the way they work and provide customers with quality products and services. In case of the manufacturing sector, digital transformation is witnessed in the way organizations use information technology to make every aspect of production more efficient, be it product design, assembly, or delivery. Implementation of Nutanix Cloud Platform built on Hyperconverged Infrastructure (HCI) by Hero MotoCorp Ltd. is a perfect example of driving digital transformation in the manufacturing sector.

Hero MotoCorp Ltd. is an Indian multinational motorcycle and scooter manufacturer headquartered in New Delhi, India. The company is one of the biggest brands in two-wheeler manufacturers in the world. As of May 27, 2021, the market capitalization of the company was ₹59,600 crore (US$8.4 billion). The company started in 1984 as a joint venture between India's Hero Group and Honda Motor Company, Japan. The motorcycles are manufactured across three benchmarked manufacturing facilities, two in Gurgaon and one in Dharuhera, located in the state of Haryana, India. In 2011, Honda agreed to transfer its entire shareholding of 26 percent of the company to the Indian Promoter Group along with the rights and licences to manufacture, assemble, sell, and distribute certain products and their services parts under their Intellectual Property Rights.

The company rolls out 1 motorbike every 18 seconds. This is possible only if Hero MotoCorp can achieve a perfect sync between the various processes, from supply chain of materials for production, to assembly of motorcycles, quality assurance, and distribution. For this, the company relies on a three-tier IT infrastructure to provide data access and connectivity so that employees can coordinate between various activities required during the production cycle. The three-tier architecture organizes applications into three logical and physical computing tiers: (1) the presentation tier or user interface; (2) the application tier, where data is processed; and (3) the data tier, where the data associated with the application is stored and managed. One of the main benefits of the three-tier architecture is that the services of each tier can be customized and optimized without impacting others.

One of the critical applications in the company is the product lifecycle management (PLM) system, which is implemented by using the three-tier infrastructure. Of late, the company identified that the system is failing and not able to deliver results, which is causing problems. The first issue identified was related to the speed at which the data was managed on the front end. This was causing hours of delay in the operations of supply chain management. For example, delays in invoicing lead to huge queues of delivery trucks at the production plants. The second issue was related to the management of IT infrastructure. It was observed that most of the time IT personnel were busy managing the IT infrastructure and could not focus on strategic work.

After much deliberation between updating the legacy system and implementing new technology to manage IT infrastructure, Mr Brahmachari and Mr Ajay Aggarwal, who run the Hero MotoCorp data centres, decided to go for HCI technology. The main reason for selecting this technology was that it combines all the elements of a traditional data centre (i.e., storage, compute, networking, and management) and provides improvement in performance, scalability, and management of IT infrastructure. Rather than existing as a separate resource, storage and compute resources are added to server hosts and managed by software. Like most modern workloads, HCI leverages virtualization technology. In sum, HCI infrastructure is managed centrally by a single piece of software that manages everything including storage, computing, or virtual machines.

Hero MotoCorp selected Nutanix HCI software and started migrating their critical applications such as PLM, vehicle dispatch system, and dealer management system to Nutanix. Thanks to Nutanix HCI software, the company observed improvement in the performance of PLM by 30 percent, resulting in faster access to the data and reducing the delays in routine tasks. Access to accurate data within time also reduced the long queues of drivers as they do not have to wait long to be attended by the personnel managing arrival and departure. Further, it also helped in the improvement of employee productivity by 20 percent as they can complete the task with better accuracy and within minutes.

The three-tier infrastructure implemented at Hero MotoCorp required IT personnel to manage separate server, storage, and networking setup which was complex and time consuming. The Nutanix HCI software provided flexible and easy-to-manage application that helped IT personnel to perform the management of IT infrastructure more efficiently. For instance, the time of scaling hardware for new services was reduced to one day and infrastructure upgrades were done in a single click. This saved a lot

UOC/5CB1677A-38B8-F111-A7E0-0050569FD662